# Supplementary Material

Assessing Resource Availability for Business Process Simulation Calibration

This supplementary material gives further detail on the approach and on its evaluation on synthetic and real-world event logs. Section S1 details the method. Section S2 describes the evaluation setup and reports the results for each dataset. The synthetic logs carry a known $P^{\text{true}}$, so the evaluation there examines which probabilities the approach revises and whether $P^{\text{cal}}$ ends closer to that setting than $P$, and it adds two sensitivity analyses. The real log has no such reference, so the evaluation there compares simulation under $P$ and under $P^{\text{cal}}$ with the observed log.

## S1 Method Details

**Estimation.** A timeslot is positive when $\rho^{\text{obs}}_{rs,\tau}$ exceeds $p_{rs,\tau}$, which is the paper's $Y^{+}_{rs,\tau} = 1$. Each aggregation group is described by six features, all computed from the observed log and the encoded probabilities $p_{rs,\tau}$. "How strongly and consistently the observed starts exceed the encoded availability" becomes three features, the mean number of occurrences with spare capacity over the group's timeslots, that same mean over the whole resource, and the share of the days of the week on which the group's timeslots are positive. "The nearby positive labels and those for the same resource and weekday" becomes two, the number of positive hours neighbouring the group's timeslots and the share of the resource's timeslots that are positive on the group's day. "The resource's overall activity level" becomes the sixth, its total observed start fraction. $\kappa^{+}$ is the share of groups with an observed positive label, and the classifier is a CatBoost model at 180 iterations, depth 4 and learning rate 0.05, with balanced class weights, fitted on the groups of all assessed settings at once. $Q^{+}$ is the score it then assigns to those same groups, ranking the groups of the log at hand rather than predicting unseen ones.

**Diagnosis.** Alongside $Q^{+}$, the rule uses the difference $\Delta_{rs,\tau} = \rho^{\text{obs}}_{rs,\tau} - \rho^{\text{sim}}_{rs,\tau}$. Its sign gives the direction of a revision, and its group mean $\bar{\Delta}_{rs,\tau}$ determines whether a decrease is allowed. The thresholds that gate the two directions come from group quartiles, $\theta^{+}$ marking the top quarter by $Q^{+}$, $\theta^{-}_{Q}$ the bottom quarter by $Q^{+}$, and $\theta^{-}_{\Delta}$ the most negative quarter of the negative group mean differences. All three are taken over the groups of all assessed settings together, at the same shares on both datasets.

**Calibration.** The classifier is fitted once per seed and the rule is applied to the scores of that one fit. A revision uses $Q^{+}$ twice. First, which groups are revised is a cut at the top quarter, so it turns on how well the classifier separates the understated groups from the rest, which an area under the ROC curve measures. Second, an increase moves a probability by $Q^{+}_{rs,\tau}\Delta_{rs,\tau}$, so its size rests on the score and the difference together. The distance to $P^{\text{true}}$ covers both, and that is the measure the paper reports. None of the values named in this section changes between the two datasets.

## S2 Evaluation

### S2.1 Setup

Both evaluations follow the same protocol, and the two datasets differ in two places only, the width of the time blocks and the pooling of weekdays. Extraction measures both start fractions on the training weeks, so that $\rho^{\text{obs}}_{rs,\tau}$ and $\rho^{\text{sim}}_{rs,\tau}$ cover the same span. Each log contributes four training weeks, and the distances are measured on the four test weeks that follow. Estimation then collects the timeslots into aggregation groups, one per resource, day class and block of hours. The real log is sparser, so its blocks are wider, four hours, 07:00–11:00, 11:00–15:00, 15:00–19:00, against two on the synthetic side, 08:00–10:00, 10:00–12:00, 12:00–14:00, 14:00–16:00. The hours outside those blocks form one further group per resource and day. $P^{\text{true}}$ is the same on every working day, so the synthetic side pools the five weekdays into one class and the weekend into another. The real side estimates per weekday instead.

A calibrated setting is judged by how far the log it simulates is from the observed one, and three Wasserstein distances provide that measure. CWD and CED compare the hourly distributions of active resources and of recorded events, CTD the case duration distributions. All three work on empirical distributions, so the number of events behind a distribution does not affect the distance. Each calibrated setting is re-simulated five times per seed, and each run is scored separately against the test weeks. The improvement $(D_{\text{before}} - D_{\text{after}})/D_{\text{before}}$ is formed inside one re-simulation. Each seed is summarised by

its median over those five, and the tables report the median of the ten seed values, followed after ± by half the interquartile range across the seeds. The count in brackets is separate from both, and gives how many of the ten seeds improved at all. A positive value means the distance decreased under $P^{\text{cal}}$.

### S2.2 Synthetic Evaluation

**Dataset.** The synthetic process is a sequence of two activities served by five resources that the model treats as identical. They work only between 08:00 and 16:00, Monday to Friday. Figure S1 shows the model together with its simulation inputs, the arrival rate and the processing time. Each log covers eight weeks. The arrival rate is two per hour, and the sensitivity analyses at the end of this section vary it together with how far a setting departs from $P^{\text{true}}$. The model's other inputs are fixed throughout. The input under study is the availability setting. $P^{\text{true}}$ is the one the resources follow when the observed log is produced, and Table S1 gives the seven settings assessed against it. Figure S2 draws them over a working day. Its third panel is the reported run, with three of the five resources deviating, and the other panels vary that number for the sensitivity analysis below. A setting's observed log is one run of the model under $P^{\text{true}}$, the same log for all ten of its seeds.

**Table S1.** The assessed availability settings

| Setting | Type | Workload | Change against $P^{\text{true}}$ |
|---|---|---|---|
| $P_{\text{S0-same}}$ | control, no deviation | baseline | $P$ equals $P^{\text{true}}$. |
| $P_{\text{S1-low}}$ | availability too low | baseline | Availability of CR01, CR02, CR03 halved. The other resources equal $P^{\text{true}}$. |
| $P_{\text{S1-high}}$ | availability too high | baseline | Availability of CR01, CR02, CR03 raised by half and clipped at 1. The other resources equal $P^{\text{true}}$. |
| $P_{\text{S2-early}}$ | shifted earlier | baseline | Each resource's profile inside 08:00-16:00 moved two hours earlier. The two hours it vacates at the end of the day keep the profile's lowest value. |
| $P_{\text{S2-late}}$ | shifted later | baseline | Each resource's profile inside 08:00-16:00 moved two hours later. The two hours it vacates at the start of the day keep the profile's lowest value. |
| $P_{\text{S3-busy}}$ | control, more work | high | Arrival rate scaled by 1.5. Availability equals $P^{\text{true}}$. |
| $P_{\text{S3-idle}}$ | control, less work | low | Arrival rate scaled by 0.5. Availability equals $P^{\text{true}}$. |

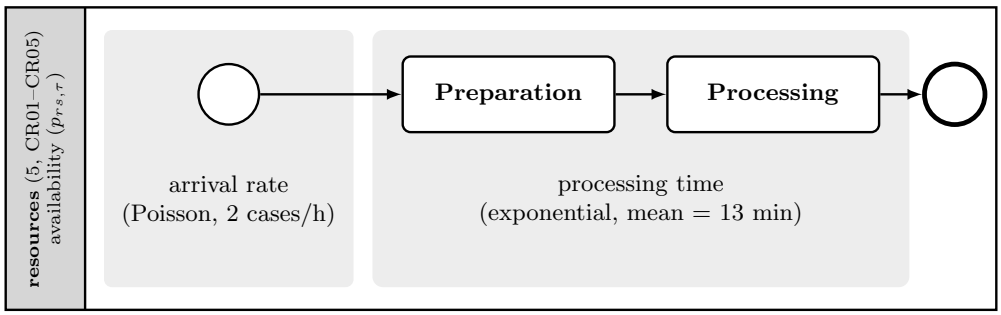



**Fig. S1.** The simulation model and the inputs that the settings change

**Results.** The revisions concentrate on the resources whose availability deviates from $P^{\text{true}}$ and move in the direction the deviation calls for, which is the result the paper reports. On the S1 pair about three quarters of them fall on CR01–CR03, all upward where availability was set too low and all downward where it was set too high, and the rest land on the two resources that were already correct. In the S2 settings every resource deviates, so the revisions carry no information about placement there, and both directions occur. Two of the three controls are revised as well, $P_{\text{S0-same}}$ and $P_{\text{S3-busy}}$ only upward. The comparison therefore does react to workload alone, although only to an increase in it. $P_{\text{S3-idle}}$ is not revised at all. Over the groups a deviation can reach, $Q^{+}$ ranks the understated ones with an area under the ROC curve of 0.857, and 0.875 on those that carry no positive label of their own. The group mean

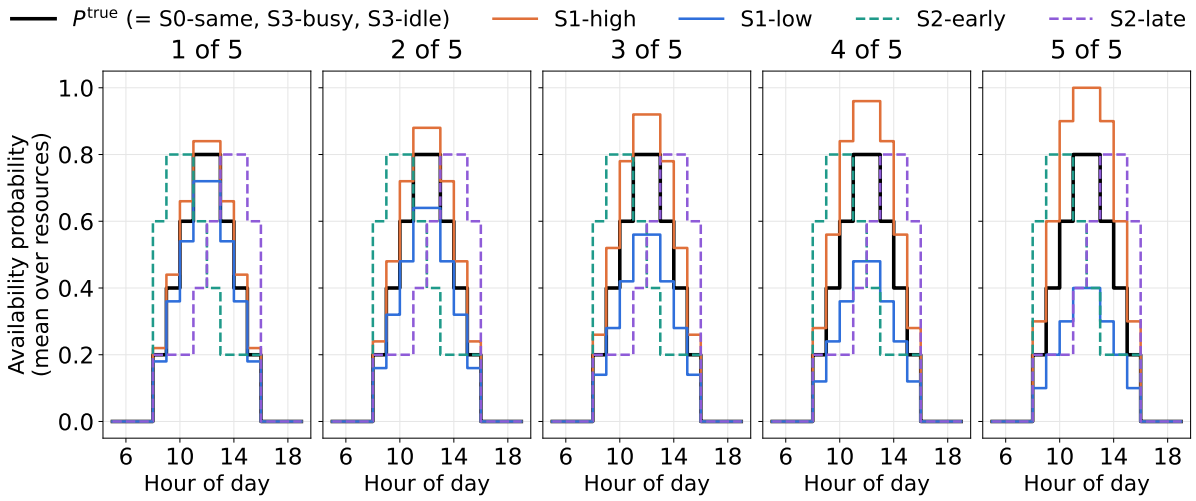



**Fig. S2.** The assessed settings at each number of deviating resources

difference and the count of neighbouring positive hours do as well on their own, so the order rests on the labels and the features rather than on the fitted combination.

Table S2 gives what the revisions were worth. Its first three columns give the mean absolute difference to $P^{\text{true}}$ under $P$, $P^{\text{cal}}$ and $P^{\text{emp}}$. $P^{\text{cal}}$ is the closest of the three on the S1 pair, in every seed, because $P$ is already exact on the resources that do not deviate while the observed fractions that $P^{\text{emp}}$ copies are off by 0.139 there. On the S2 pair every resource deviates and no correct one is left to disturb, so $P^{\text{emp}}$ is the closer, again in every seed. Compared with $P$, it is the closer on all four deviation settings and in every seed.

Below the line in the table $P$ equals $P^{\text{true}}$ exactly, so the $P^{\text{cal}}$ and $P^{\text{emp}}$ columns measure error that calibration and $P^{\text{emp}}$ introduce rather than remove. The three controls differ in nothing but the workload, and the table orders them by it. The spurious drift grows with the workload, because the rules fire upward and more work puts more observed starts above the encoded probability. $P^{\text{emp}}$ moves the other way, worst on the idlest control, since a thinner log leaves fewer starts to copy. Even on the busiest control the drift stays below half of the deviation injected into $P_{\text{S1-high}}$. $P_{\text{S3-idle}}$ is never revised, so $P^{\text{cal}}$ is still $P$ there and its three behavioral improvements are zero.

The CWD, CED and CTD columns report the behavioral distances after re-simulation, as improvements over $P$. Figure S3 gives the distances themselves. CWD improves in all four deviation settings, several times more on the S2 pair than on the S1 pair. Moving availability in time changes when the work falls, which is what CWD measures. Halving or raising availability scales a resource's activity by the same factor at every hour, which leaves the daily pattern largely unchanged. CED improves in all four, again most on the S2 pair. The CTD percentages carry little information in any of the four, because the distance before calibration already sits at the level two runs of $P^{\text{true}}$ itself differ by. Across the four, calibration moves $P$ closer to $P^{\text{true}}$ wherever a deviation was injected, the behavior follows it most where the deviation moves the daily pattern, and a setting that is already correct still drifts, by less than the deviation it would have to be mistaken for.

**Sensitivity.** The results above come from one configuration, three deviating resources at the 1× arrival rate. Whether those results carry beyond that configuration depends on how large the deviation is and on how much evidence the log holds. Below, one to five of the five resources deviate in turn, and the arrival rate is scaled from 0.25 to 1.25 times the modelled one. The rate matters in two opposite ways. A higher one drives $\rho^{\text{obs}}$ towards $P^{\text{true}}$, to the point where copying the observed fractions, as $P^{\text{emp}}$ does, is already close to exact. A lower one leaves too little work to tell the re-simulated behavior from the variation a single process shows across its own runs. The reported rate sits between them. Neither analysis can be read as the varied factor acting alone. $Q^{+}$ and the thresholds come from the groups of all the settings together, so changing some settings moves the results of others. Across the five numbers of deviating resources, $P_{\text{S2-late}}$ varies by nine percentage points in its improvement over $P$, although its configuration is identical in all of them.

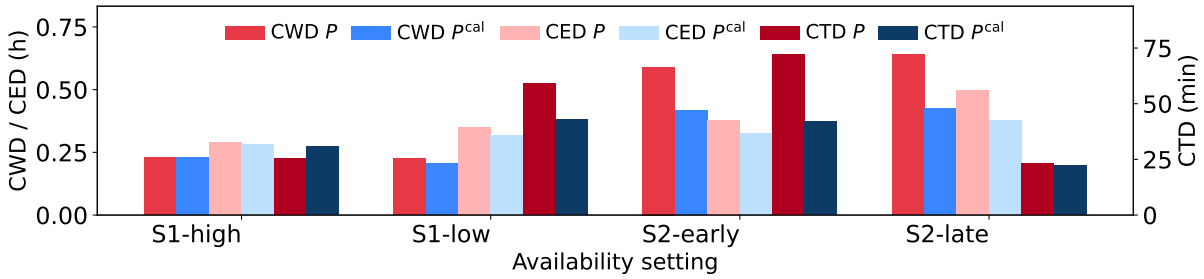



**Fig. S3.** Distances to the observed log after re-simulation

**Table S2.** Synthetic results

| | Mean $\lvert p - p^{\text{true}} \rvert$ over a week | | | Improvement in behavior | | |
|---|---|---|---|---|---|---|
| Setting | $P$ | $P^{\text{cal}}$ | $P^{\text{emp}}$ | CWD | CED | CTD |
| $P_{\text{S1-low}}$ | 0.036 | 0.022 | 0.034 | 9.2% ±5.4 (8/10) | 9.1% ±7.1 (7/10) | 35.5% ±9.7 (9/10) |
| $P_{\text{S1-high}}$ | 0.029 | 0.022 | 0.032 | 3.7% ±7.6 (6/10) | 8.4% ±10.3 (6/10) | -20.8% ±7.2 (2/10) |
| $P_{\text{S2-early}}$ | 0.065 | 0.042 | 0.027 | 29.6% ±1.7 (10/10) | 13.7% ±6.4 (9/10) | 45.4% ±18.8 (9/10) |
| $P_{\text{S2-late}}$ | 0.065 | 0.046 | 0.039 | 34.7% ±1.7 (10/10) | 25.6% ±5.9 (10/10) | 6.6% ±14.2 (7/10) |
| $P_{\text{S3-busy}}$ | | 0.007 | 0.020 | -9.0% ±4.8 (2/10) | -5.8% ±8.6 (4/10) | 2.2% ±35.0 (5/10) |
| $P_{\text{S0-same}}$ | $P = P^{\text{true}}$ | 0.005 | 0.028 | -4.1% ±12.1 (5/10) | -3.4% ±7.2 (5/10) | -3.3% ±25.5 (5/10) |
| $P_{\text{S3-idle}}$ | | 0.000 | 0.061 | 0% | 0% | 0% |

Only the S1 settings change as the number of deviating resources varies, panel by panel in Figure S2. The other settings are unchanged. Figure S4 then gives the revisions, the distances to $P^{\text{true}}$ and the distances after re-simulation at each number, y-axes shared along each row. The two timing shifts stayed closer to $P^{\text{true}}$ than $P$ at every number, by 20 to 36%. The S1 settings did not. With one of the five resources deviating, $P^{\text{cal}}$ ended further from $P^{\text{true}}$ than $P$ was, by 19% at worst. From two upwards it was closer in every seed and setting. A deviation therefore has to cover more than one of the five resources before the approach can resolve it.

Availability stays as it is while the arrival rate varies. The lower the rate goes, the less of $P^{\text{true}}$ the observed start fractions reach, which Figure S5 draws over a working day at each rate. At a quarter of the modelled rate every setting has working-week timeslots that record no start at all. The improvement over $P$ follows the same pattern, drawn at each rate in Figure S6, and the decline is not monotone: $P_{\text{S1-high}}$ improves more at half the rate than at the modelled one, and at three quarters it ends further from $P^{\text{true}}$ than $P$. These runs support the limitation the paper states, that the approach can only infer availability where the workload gives enough opportunities to observe it.

### S2.3 Real-World Evaluation

**Dataset.** The real process is the loan application process behind the BPI Challenge 2012 log. The log records start and completion timestamps and resource labels over 30,502 events, and it carries no record of availability. With the method of López-Pintado et al. we discovered a simulation model from the log, which provides availability calendars along with the remaining simulation inputs. Three calendars are assessed, $P_1$, $P_2$ and $P_3$, from runs that differ in one hyperparameter, which that method calls the angle, at 0.55, 0.71 and 0.97. A larger angle spreads availability over more hourly timeslots, each with a lower probability. Where the resource field is empty the log carries the literal `NOT_SET`, which the discovery treats as a resource of its own and gives a calendar. We assess and calibrate it like the rest. Calibration uses weeks 2012W02–2012W05 and the reported distances come from the four weeks that follow, held out from it.

**Results.** Table S3 reports the distances on the test weeks, summarised as in Table S2 and followed, after the semicolon, by the re-simulations that improved out of the fifty a calendar has, ten seeds by five. Simulation under $P^{\text{cal}}$ reproduced the observed daily pattern of work more closely than simulation under

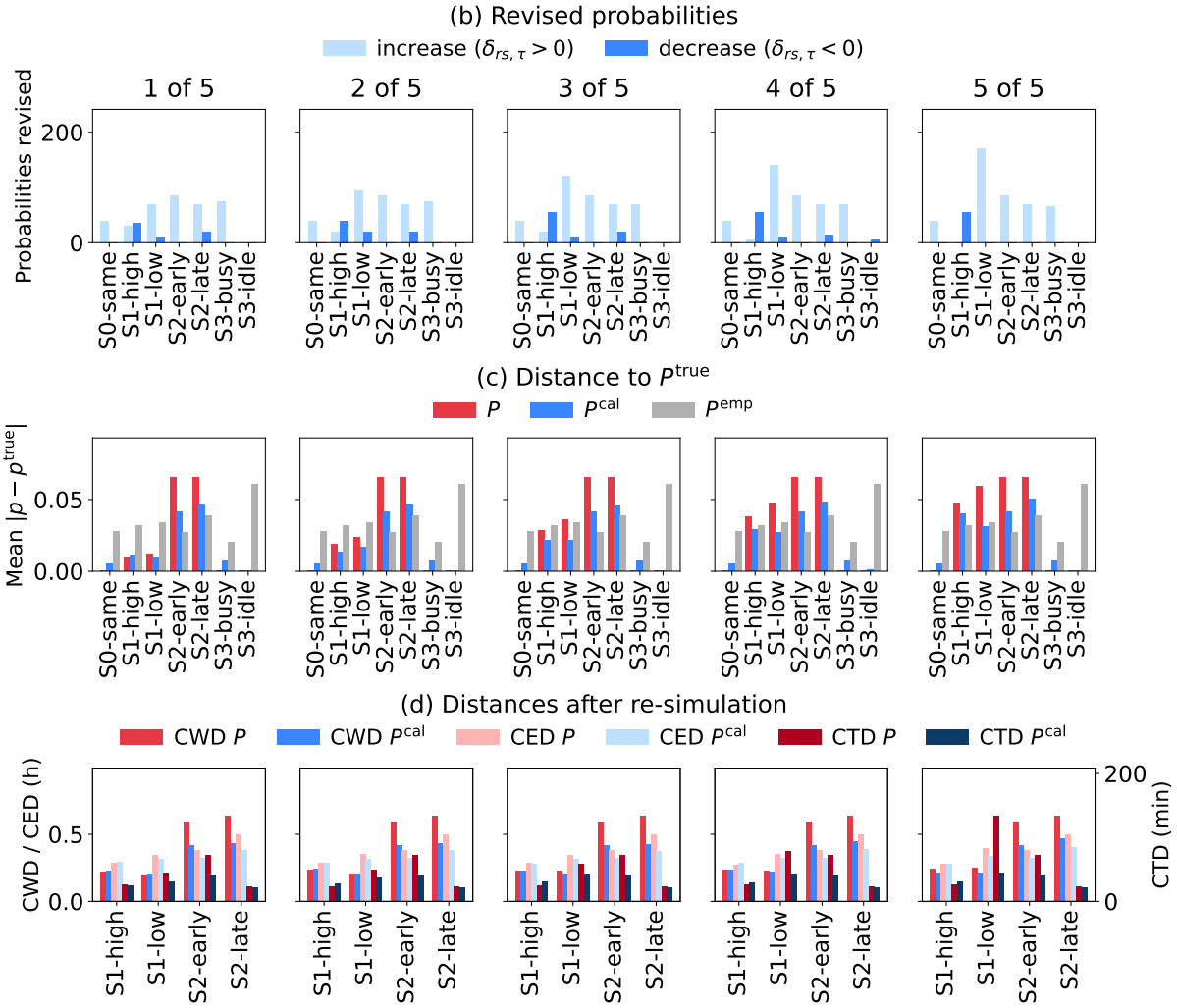



**Fig. S4.** Sensitivity to the number of deviating resources

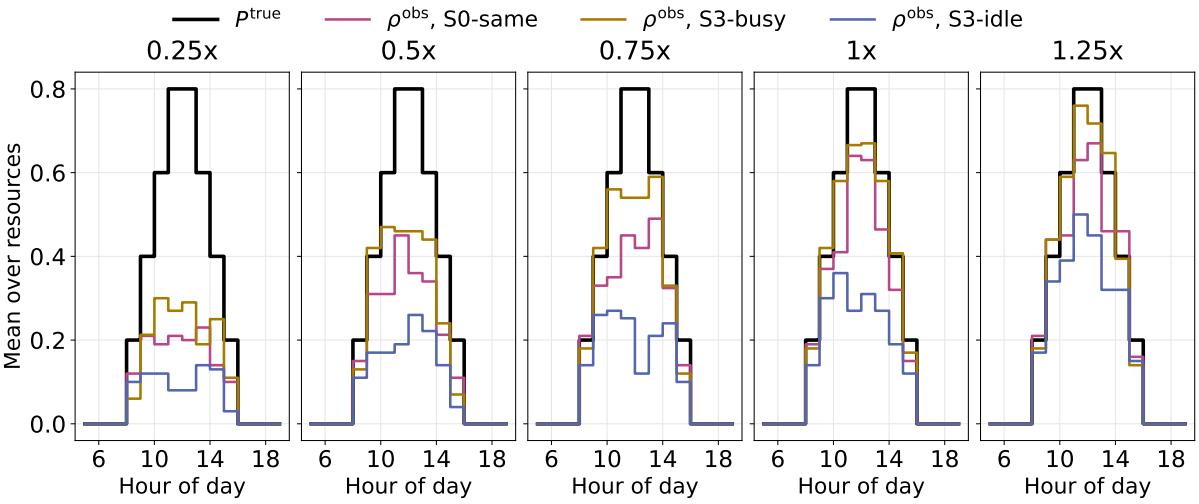



**Fig. S5.** Observed start fractions at each arrival rate

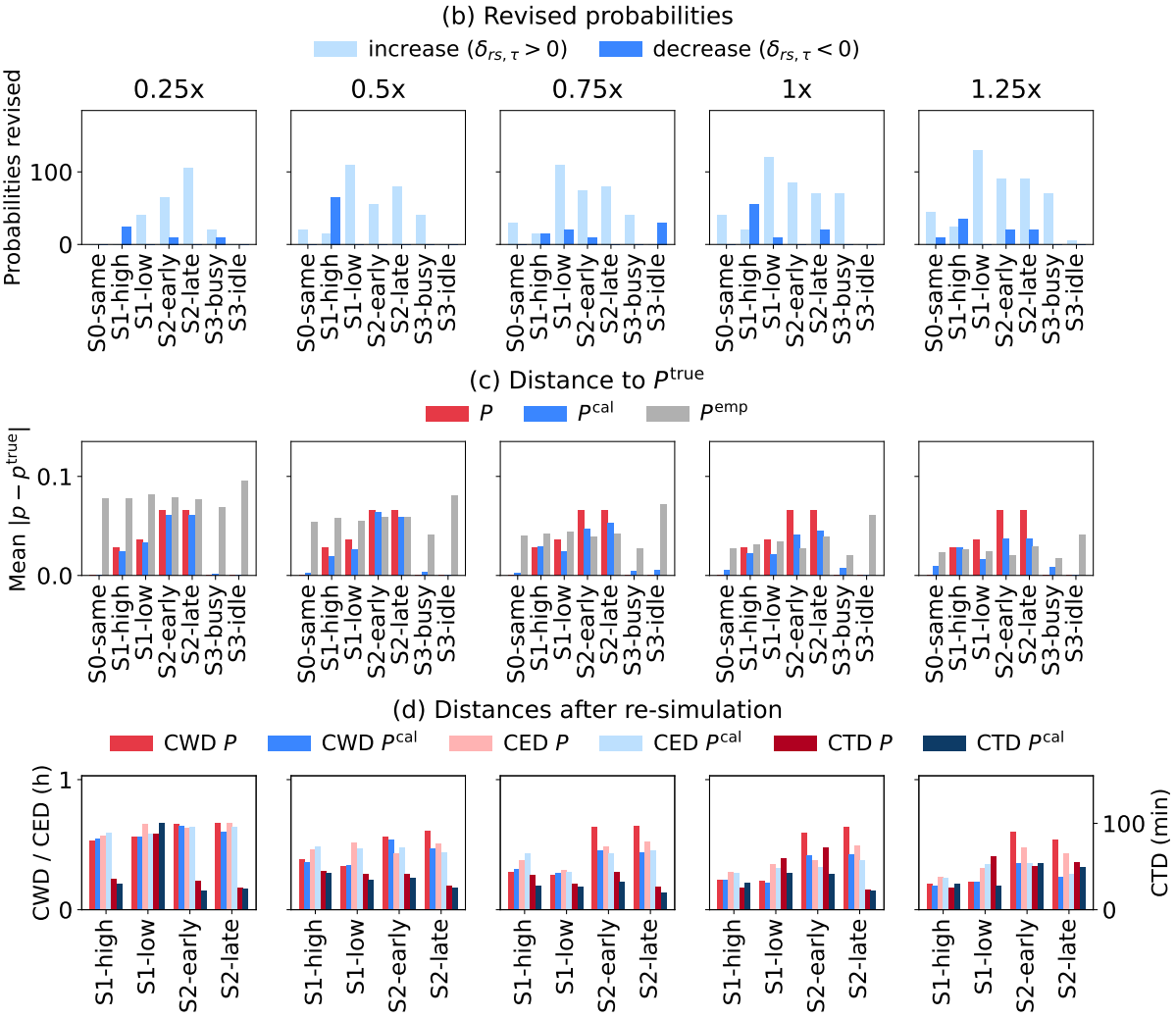



**Fig. S6.** Sensitivity to the workload

$P$. CWD and CED fell for every calendar, while CTD moved by less than a percent. Calibration also narrowed the gap on the training weeks, in every seed, by 10.9 to 14.4% in the median seed of a calendar and by at least 9.6% in the weakest. Nearly all of the revisions it made there were increases, 90.6 to 94.1%, and about a sixth moved the start fraction away from $\rho^{\text{obs}}$ rather than towards it. The discovered models fall far short on case duration, the observed median running some fifty times the simulated one, which sets the scale of the CTD results. Calibration moves it by less than an hour. Figure S7 has the start fractions, the revisions and the distances for the calendars. Calibration therefore improves the daily pattern of work, which is what its revisions act on, and does so for all three calendars.

**Table S3.** Real-world results on the test weeks

| Setting | CWD | CED | CTD |
|---|---|---|---|
| $P_1$ | 4.7% ±0.9 (10/10; 44/50) | 11.4% ±2.2 (10/10; 50/50) | -0.6% ±0.2 (0/10; 10/50) |
| $P_2$ | 11.0% ±0.9 (10/10; 49/50) | 21.5% ±1.9 (10/10; 50/50) | -0.6% ±0.3 (2/10; 16/50) |
| $P_3$ | 12.8% ±1.2 (10/10; 50/50) | 16.3% ±1.9 (10/10; 50/50) | -0.4% ±0.1 (1/10; 16/50) |

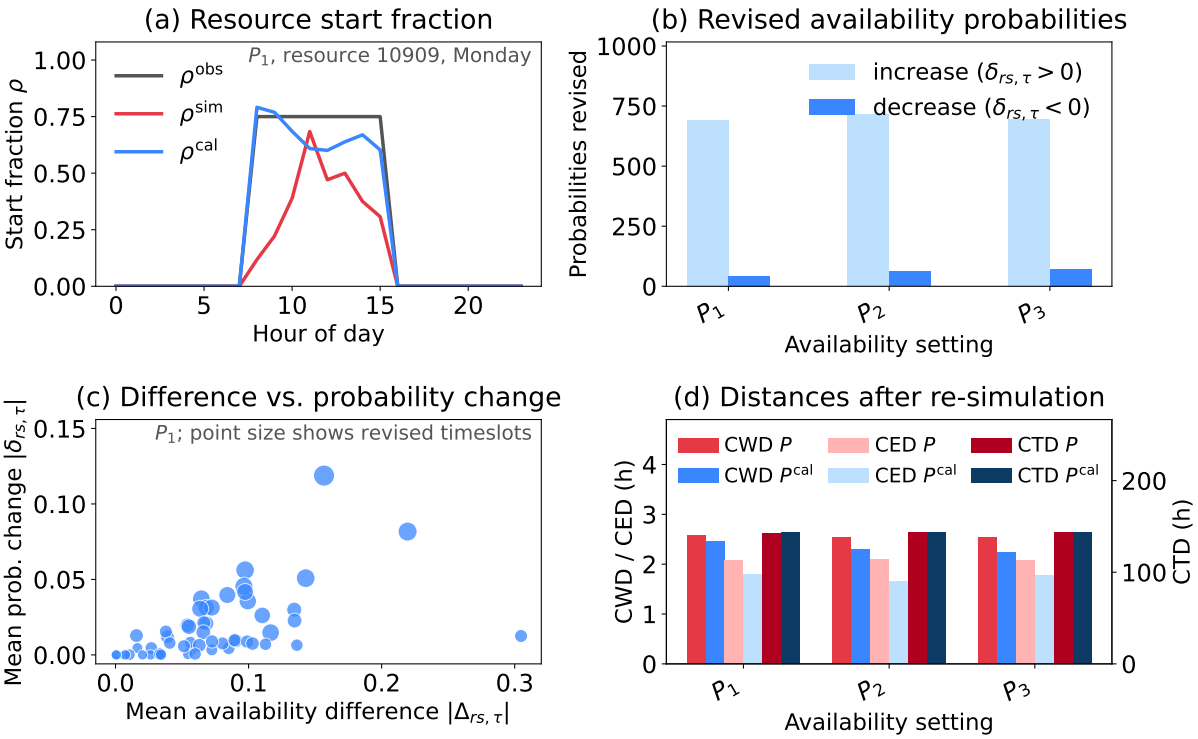



**Fig. S7.** Real-world evaluation